Tissue-level associations between macrophage efferocytosis-related programs and vascular smooth muscle cell state signatures in symptomatic versus asymptomatic human carotid plaques

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Abstract

Background: Defective macrophage efferocytosis and vascular smooth muscle cell (VSMC) state switching are implicated in atherosclerotic plaque biology, but their relationship in clinically defined human carotid plaques remains incompletely resolved. Methods: We analyzed legally accessible public human carotid plaque transcriptomic datasets from GEO: a discovery Affymetrix microarray cohort, GSE111782 (9 symptomatic and 9 asymptomatic plaques), an independent diabetic RNA-seq validation cohort, GSE311535 (6 symptomatic and 6 asymptomatic plaques), and a Smart-seq2 single-cell cohort, GSE260657 (8 symptomatic and 7 asymptomatic carotid plaque samples; 7,690 cells after parsing). Predefined efferocytosis and VSMC state signatures were evaluated using assay-appropriate bulk models, rank-based enrichment, sample-level scoring, Spearman correlation, and donor-level single-cell summaries. P values were two-sided where applicable and adjusted using the Benjamini-Hochberg method. Results: No individual genes reached FDR < 0.05 in the discovery or validation bulk differential-expression analyses, and no predefined sample-level signature differed between symptomatic and asymptomatic plaques at FDR < 0.05. In the discovery cohort, ranked signature enrichment showed lower-ranked efferocytosis and VSMC programs in symptomatic plaques, but this pattern was not reproduced at FDR < 0.05 in the validation cohort. Across bulk samples, efferocytosis scores were positively associated with inflammatory VSMC scores in both GSE111782 (Spearman rho = 0.800, FDR = 0.0014) and GSE311535 (rho = 0.881, FDR = 0.0015). Single-cell analysis localized macrophage and VSMC compartments, but donor-level macrophage or VSMC program comparisons did not reach FDR < 0.05. Conclusion: The most reproducible observation was a tissue-level association between efferocytosis-related and inflammatory VSMC transcriptional programs rather than a robust symptomatic-versus-asymptomatic differential signature. These public-data results identify candidate associations for experimental follow-up and do not establish causality, pathology-defined plaque status, diagnostic utility, or therapeutic actionability.

Keywords: carotid atherosclerosis; symptomatic plaque; asymptomatic plaque; efferocytosis; macrophage; vascular smooth muscle cell; transcriptomics

Introduction

Human carotid atherosclerotic plaques are clinically important because symptomatic plaques are associated with cerebrovascular events, yet symptom status is not identical to a pathology-defined plaque category. This distinction matters for transcriptomic studies: datasets that label samples as symptomatic versus asymptomatic should not be rewritten using pathology-defined categories unless the original metadata or linked pathology report supports that terminology. The present study therefore retains the clinical labels reported in each dataset.

Macrophage handling of apoptotic cells is one candidate process linking inflammation resolution, necrotic-core formation, and advanced plaque biology. Efferocytosis requires receptors, bridging molecules, phagolysosomal processing, and lipid-handling responses, and prior experimental and human evidence has linked defective efferocytosis with atherosclerotic inflammation and plaque necrosis [6-8]. In parallel, VSMCs in atherosclerotic plaques occupy contractile and non-contractile states, including matrix-producing, inflammatory, osteogenic, and macrophage-like programs [9-13]. These state changes complicate bulk tissue interpretation because cell composition and cell-state activity can both influence transcriptomic signals.

Single-cell studies have sharpened this problem by showing substantial cellular heterogeneity within human atherosclerotic plaques, including macrophage state diversity and VSMC phenotypic modulation [12-20]. However, less is known about whether a predefined macrophage efferocytosis-related transcriptional program is reproducibly associated with VSMC state signatures in clinical symptomatic versus asymptomatic human carotid plaque cohorts.

Here we performed a secondary analysis of public human carotid plaque datasets to test a bounded hypothesis: symptomatic plaques would show altered efferocytosis-related transcriptional features and these features would be associated with non-contractile VSMC programs. We used one discovery bulk cohort, one independent bulk validation cohort, and one human carotid plaque single-cell cohort, with predefined gene sets and donor-aware single-cell summaries. The analysis was designed to identify reproducible associations, not to infer causality or reclassify symptomatic plaques using pathology-defined categories.

Materials and Methods

Data sources and cohort definitions

Public datasets were identified from GEO and retained only when the metadata supported human carotid atherosclerotic plaque tissue and clinically interpretable grouping. GSE111782 was used as the discovery bulk cohort because its GEO record and linked publication describe post-bifurcation internal carotid atheroma samples with symptomatic and asymptomatic clinical labels [1,4,5]. This cohort contained 18 samples, with 9 symptomatic and 9 asymptomatic plaques, profiled on the GPL571 Affymetrix Human Genome U133A 2.0 Array. GSE311535 was used as the independent bulk validation cohort because it contains human carotid plaque RNA-seq count data from patients with diabetes, including 6 symptomatic and 6 asymptomatic samples [2]. GSE260657 was used for single-cell localization because the GEO record and linked publication describe Smart-seq2 data from 15 human carotid plaque samples, including 8 symptomatic and 7 asymptomatic samples [3]. Dataset provenance, URLs, limitations, and audit decisions are recorded in DATA_PROVENANCE.md and 03_data/metadata/dataset_audit.csv.

Predefined gene sets and signatures

The efferocytosis gene set was defined before inspecting project results and included receptors, bridging molecules, phagocytic/lysosomal genes, and lipid-handling genes relevant to apoptotic-cell recognition, engulfment, and processing. Source bases included Gene Ontology, Reactome, UniProt annotations, and published atherosclerosis/efferocytosis literature [6-8,27,28]. VSMC state signatures were likewise predefined and represented contractile VSMC, synthetic VSMC, inflammatory VSMC, and ECM-remodeling/osteogenic VSMC programs, based on canonical markers and prior plaque single-cell studies [9-13]. These signatures were treated as expression programs rather than definitive lineage states.

Bulk transcriptomic processing

GSE111782 was analyzed as an RMA-processed log2 microarray expression matrix. Probes without usable gene annotation were removed where possible, duplicate gene mappings were handled by retaining the probe with the highest average expression, and the symptomatic-versus-asymptomatic contrast was fitted using limma empirical Bayes linear modeling [21]. GSE311535 was analyzed as gene-level RNA-seq counts using edgeR quasi-likelihood negative-binomial models after low-expression filtering and library-size normalization [22]. In both cohorts, positive log2 fold change denotes higher expression in symptomatic plaques relative to asymptomatic plaques. Differential-expression P values were adjusted using the Benjamini-Hochberg method, with FDR < 0.05 as the prespecified threshold.

Signature, enrichment, and association analyses

Sample-level signature scores were computed from predefined gene sets using available mapped genes; the score represents the mean standardized expression of genes available on the relevant platform. Group comparisons used two-sided Wilcoxon rank-sum tests because the bulk cohorts were small and no normality assumption was defensible. Ranked enrichment of predefined signatures used preranked symptomatic-versus-asymptomatic statistics following the GSEA framework [24]. GO biological process, KEGG, and Reactome over-representation analyses were attempted using clusterProfiler and ReactomePA when gene mapping and package availability allowed [25-29]. All enrichment and signature tests used Benjamini-Hochberg correction. Tissue-level associations between signatures were evaluated using Spearman correlations. WGCNA was prespecified but not run because the discovery bulk sample size was 18, below the project threshold of at least 30 samples and therefore not adequate for stable network inference [30].

Single-cell processing and annotation

GSE260657 raw Smart-seq2 text files were parsed without modifying the original downloaded archive. Cells were evaluated using prespecified quality-control rules for detected genes, counts, mitochondrial fraction, and sample-level parsing. Because the available workflow was a lightweight reproducible fallback rather than a full Seurat/Scanpy integration, cell annotation used canonical marker scoring, PCA/k-means clustering, and marker-candidate inspection. Major cell classes included macrophage, VSMC, endothelial cell, fibroblast, T/NK cell, B cell, mast cell, and unassigned. Uncertain cells were left as unassigned. Group comparisons were performed at donor/sample level when possible; individual cells were not treated as independent patients.

Reporting, software, and reproducibility

Analyses were performed in R 4.6.0 using limma 3.68.5, edgeR 4.10.4, GSVA 2.6.6, fgsea 1.38.0, clusterProfiler 4.20.0, ReactomePA, ggplot2 4.0.3, and related Bioconductor packages; package versions are archived in 09_environment/package_versions.csv and 09_environment/session_info.txt. Scripts are stored in 04_scripts, logs in 08_logs, and generated tables and figures in 05_results. No samples were removed only because they changed the result direction or significance.

Results

Public cohort selection produced discovery, validation, and single-cell evidence layers

The dataset audit retained GSE111782 as the discovery bulk cohort, GSE311535 as an independent bulk validation cohort, and GSE260657 as the single-cell localization cohort (Figure 1). The bulk cohorts were not merged because they differed by platform, preprocessing, and clinical context; GSE311535 is diabetes-specific. The single-cell cohort passed the project gate for human carotid plaque localization after raw file parsing, donor/sample metadata review, and marker-based annotation. Optional spatial transcriptomics, cell-cell communication, and machine-learning modules were not used in the main conclusions because their strict data, annotation, and overfitting-control gates were not met.

Bulk differential-expression analysis did not identify FDR-significant individual genes

In GSE111782, limma tested 14,289 genes and identified 0 genes at FDR < 0.05 (Figure 2). The smallest nominal P-value genes were SLC35F5 (log2FC = -2.068, P = 4.53e-04, FDR = 0.624), SOX5 (log2FC = -1.287, P = 0.0011, FDR = 0.624), ATP2B2 (log2FC = 1.444, P = 0.0022, FDR = 0.624), but their adjusted FDR values were not below the prespecified threshold. In GSE311535, edgeR tested 21,925 genes and identified 0 genes at FDR < 0.05. The smallest nominal P-value genes were PI4KAP1 (log2FC = 1.308, P = 8.18e-05, FDR = 0.951), CTC-462L7.1 (log2FC = -0.480, P = 9.68e-04, FDR = 0.951), AC010886.2 (log2FC = -0.690, P = 0.0012, FDR = 0.951), again without FDR support. Across 10,864 genes available in both bulk cohorts, 6,609 (60.8%) had concordant symptomatic-versus-asymptomatic log2FC directions, but no gene reached FDR < 0.05 in both cohorts. These results argue against presenting individual DEGs as validated markers in the current public-data analysis.

Discovery ranked enrichment suggested lower efferocytosis and VSMC program ranks in symptomatic plaques, but validation was weaker

Ranked enrichment of predefined signatures in the discovery cohort identified FDR-supported negative enrichment for contractile_vsmc (NES = -1.987, FDR = 0.010); efferocytosis (NES = -1.601, FDR = 0.026); inflammatory_vsmc (NES = -1.809, FDR = 0.026); synthetic_vsmc (NES = -1.841, FDR = 0.026). Because positive log2FC denotes higher expression in symptomatic plaques, negative normalized enrichment scores indicate that these predefined genes tended to occur lower in the symptomatic-versus-asymptomatic ranked list. GO biological process over-representation in the discovery cohort returned three FDR-supported terms related to cartilage or chondrocyte differentiation: regulation of chondrocyte differentiation (FDR = 0.035); positive regulation of chondrocyte differentiation (FDR = 0.035); regulation of cartilage development (FDR = 0.035). Reactome over-representation did not yield FDR-supported terms in the discovery output. In the validation cohort, none of the ranked predefined signatures reached FDR < 0.05; the lowest adjusted value was observed for efferocytosis (NES = 1.254, FDR = 0.170). Thus, the discovery enrichment pattern was not reproduced as a validation-level result.

Sample-level signature group differences did not survive FDR correction

At the sample-score level, no predefined efferocytosis or VSMC signature differed between symptomatic and asymptomatic plaques at FDR < 0.05 in either bulk cohort (Figure 3). In GSE111782, the efferocytosis median score was 0.374 in asymptomatic plaques and -0.021 in symptomatic plaques (P = 0.331, FDR = 0.757). The contractile VSMC score showed a nominal difference in GSE111782 (median 0.389 versus -0.029; P = 0.034), but this did not survive correction (FDR = 0.341). In GSE311535, the efferocytosis median was -0.142 in asymptomatic plaques and 0.224 in symptomatic plaques (P = 0.575, FDR = 0.822). These results were retained as negative or inconclusive findings rather than reframed as positive group differences.

Efferocytosis scores were reproducibly associated with inflammatory VSMC scores at tissue level

The strongest reproducible bulk observation was a positive tissue-level association between the efferocytosis and inflammatory VSMC signatures. In GSE111782, the efferocytosis score correlated with the inflammatory VSMC score (Spearman rho = 0.800, P = 6.78e-05, FDR = 0.0014) and with the synthetic VSMC score (rho = 0.593, P = 0.0094, FDR = 0.047). In GSE311535, the efferocytosis score again correlated with the inflammatory VSMC score (rho = 0.881, P = 1.53e-04, FDR = 0.0015), whereas the association with ECM-remodeling VSMC was weaker and did not pass FDR correction (rho = 0.483, FDR = 0.204). These correlations were analyzed across mixed plaque tissue samples and therefore cannot distinguish coordinated cell-state activity from cell-composition differences or shared inflammatory confounding.

Single-cell analysis localized macrophage and VSMC compartments without donor-level FDR-supported group differences

GSE260657 yielded 7,690 parsed cells across 15 donor/sample files. Marker-score annotation assigned major cell classes as follows: macrophage: 2271, fibroblast: 1645, vsmc: 1511, endothelial: 756, unassigned: 752, t_nk: 332, mast: 216, b_cell: 207 (Figure 4). Macrophages represented the largest annotated compartment (n = 2,271), and VSMCs were also clearly represented (n = 1,511). Secondary macrophage analysis produced three descriptive subclusters: macrophage_S1 n = 331, efferocytosis score = 0.730; macrophage_S2 n = 1184, efferocytosis score = 0.135; macrophage_S3 n = 756, efferocytosis score = 0.174 (Figure 5). Secondary VSMC analysis similarly produced three descriptive VSMC subclusters: vsmc_S1 n = 695, contractile score = 0.800; vsmc_S2 n = 519, contractile score = 1.064; vsmc_S3 n = 297, contractile score = 1.879 (Figure 6). Donor-level program comparisons across macrophage and VSMC compartments did not identify FDR-supported symptomatic-versus-asymptomatic differences (number of tested cell-type/program comparisons with FDR < 0.05: 0). The single-cell data therefore support cellular localization of the predefined programs but do not provide independent donor-level evidence for group differences.

Independent validation supported the efferocytosis-inflammatory VSMC association but not a validated classifier or marker set

The independent validation cohort confirmed the positive tissue-level association between efferocytosis and inflammatory VSMC scores, but it did not validate individual DEGs, sample-level signature group shifts, or discovery ranked enrichment at FDR < 0.05 (Figure 7). Because the discovery and validation cohorts were small and heterogeneous, and because no FDR-supported gene-level marker set was established in discovery and validation, the prespecified optional machine-learning module was skipped. The results are therefore reported as reproducible transcriptional associations rather than as a diagnostic model or treatment-target discovery.

Discussion

This secondary analysis found limited evidence for robust symptomatic-versus-asymptomatic differential expression at the individual gene or sample-level signature level. Neither bulk cohort produced FDR-significant individual genes, and no predefined efferocytosis or VSMC sample-level signature group comparison reached FDR < 0.05. These negative findings are important because the small cohorts and heterogeneous public metadata create a high risk of overinterpreting nominal P values.

The most consistent result was instead a cross-cohort tissue-level association between efferocytosis-related scores and inflammatory VSMC scores. This association was observed in both the discovery microarray cohort and the independent diabetic RNA-seq cohort. The finding is biologically plausible in light of prior work on macrophage efferocytosis, inflammatory plaque states, and VSMC phenotypic modulation [6-14], but it remains an association across mixed tissue samples. It may reflect coordinated activation of macrophage and VSMC programs, shared inflammatory burden, variation in cellular composition, or differences in lesion stage that could not be fully adjusted using the available metadata.

The single-cell cohort helped localize the relevant compartments, showing that macrophages and VSMCs were both represented in human carotid plaque data and that efferocytosis and VSMC-state programs could be scored at cell and donor levels. However, donor-level program comparisons were not FDR-significant. This constraint prevents a stronger claim that symptomatic plaques consistently contain a distinct efferocytosis-high macrophage state or a distinct non-contractile VSMC state in the analyzed single-cell data. The current single-cell results are therefore best viewed as localization and hypothesis-refinement evidence.

The discovery ranked enrichment results add nuance. In GSE111782, several predefined signatures, including efferocytosis and VSMC programs, were negatively enriched along the symptomatic-versus-asymptomatic ranked list. Because these enrichment signals were not reproduced in GSE311535, they should be treated as cohort-specific observations rather than validated disease signatures. Differences in platform, diabetes status, sample size, tissue handling, plaque stage, and clinical symptom definitions may all contribute to this non-replication.

Several limitations should be explicit before submission. First, all analyses used public retrospective datasets, and patient-level clinical covariates were limited. Second, symptom status was used exactly as reported and was not treated as synonymous with a pathology-defined plaque category. Third, bulk plaque tissue is a mixture of immune, stromal, endothelial, smooth muscle, and other cell populations, so tissue-level signature associations may be composition-driven. Fourth, the single-cell workflow used a lightweight marker-score annotation fallback, and full reanalysis with the original authors' processed objects or a dedicated Seurat/Scanpy pipeline would strengthen annotation confidence. Fifth, no spatial carotid validation, lineage tracing, perturbation experiment, or wet-lab validation was performed. These boundaries preclude causal, diagnostic, therapeutic, or mechanism-confirming claims.

In summary, the data support a conservative conclusion: public human carotid plaque transcriptomes show reproducible tissue-level covariation between macrophage efferocytosis-related and inflammatory VSMC transcriptional programs, but they do not establish robust symptomatic-versus-asymptomatic DEGs or validated group-discriminating signatures. Future work should test these candidate associations in prospectively phenotyped carotid plaque cohorts with harmonized clinical definitions, spatial validation, and experimental models that can separate cell-state changes from cell-composition effects.

Data Availability Statement

Publicly available datasets were analyzed in this study. These data are available through the Gene Expression Omnibus under accession numbers GSE111782, GSE311535, and GSE260657. Accession-level URLs, download dates, file names, processing scripts, generated outputs, uses, and limitations are recorded in DATA_PROVENANCE.md and 03_data/metadata/dataset_audit.csv.

Code Availability Statement

Analysis scripts, parameters, logs, intermediate tables, generated figures, and manuscript-generation scripts are included in this project directory. Before journal submission, the authors should archive the code in a persistent public repository and replace this placeholder with the repository URL: [[CODE REPOSITORY URL]].

Ethics Statement

This manuscript reports a secondary analysis of public, de-identified human datasets. The original studies' ethics approval and consent statements must be verified from the corresponding source publications before submission: [[ETHICS STATEMENT TO BE VERIFIED]].

Author Contributions

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The authors should disclose any generative AI assistance in accordance with Frontiers policy. This project used AI assistance for code orchestration, manuscript organization, and language revision; quantitative analyses and figures were generated from public data using the recorded scripts. The final wording of this disclosure must be verified by the submitting authors.

Contribution to the Field

Human carotid plaque transcriptomic studies often compare symptomatic and asymptomatic plaques, but symptom status, pathology-defined plaque categories, and mechanistic causality are sometimes conflated. This study provides a reproducible public-data analysis focused on a specific biological question: whether macrophage efferocytosis-related transcriptional features are associated with VSMC state programs in clinically defined human carotid plaques. By keeping discovery, validation, and single-cell localization analyses separate, the work identifies a conservative and reproducible signal: efferocytosis-related scores covary with inflammatory VSMC scores at the tissue level in two independent bulk cohorts. At the same time, the analysis shows that individual DEGs and sample-level signature group differences are not FDR-supported in the available cohorts. The accompanying single-cell analysis localizes macrophage and VSMC compartments but does not treat thousands of cells as independent patients. The study therefore offers a transparent candidate association for future experimental and spatial validation, while explicitly avoiding overstatement as a causal mechanism, pathology-defined signature, diagnostic tool, or therapeutic target.

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